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Faculty of Science

Computer Networks (2025-2026)

Part 1: Computer Networks and the Internet

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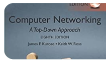
Table of contents

History of the Internet

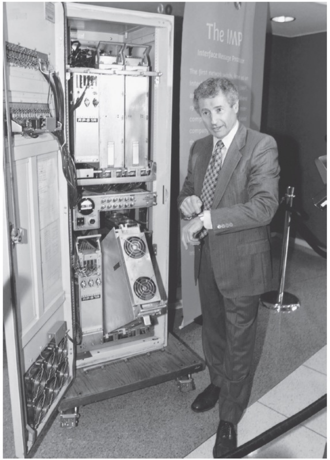
Link to the book shown at start of each section



Section 1.7

1961 – Kleinrock theoretically proves effectiveness of packet-switching

- At the time, telephone networks made use of circuit switching, which works well for constant rate voice
- A solution was needed for transmitting computer data, which is more bursty



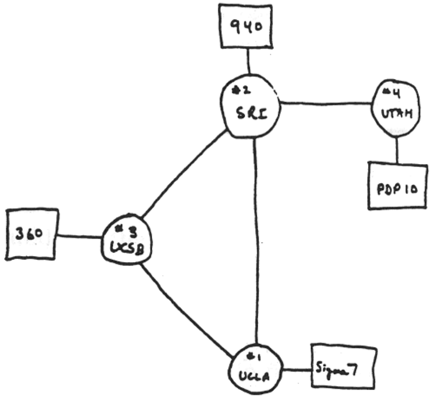
Mark J. Terrill/AP Photo

early packet switch

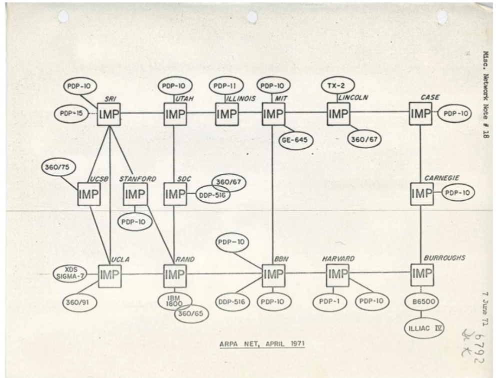
Leonard Kleinrock (MIT)

1967 – ARPAnet is conceived, the predecessor of the modern Internet

1969 (4 nodes)



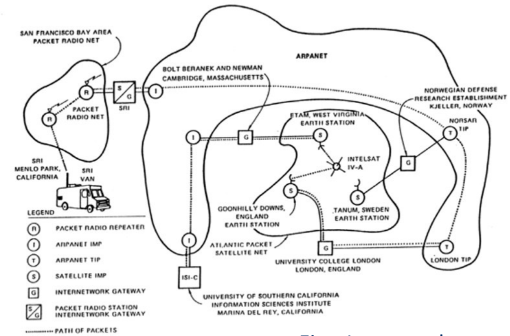
1971 (15 nodes)



- First host-to-host protocol: Network Control Protocol (NCP)
- NCP enables applications (e.g., e-mail)

1974 – Network of networks proposed by Vint Cerf and Robert Kahn

- In the early and mid 1970's many packet switched networks, besides ARPAnet, were deployed
- In 1974, Cerf and Kahn proposed an architecture to interconnect them ("internetworking")
- Their proposal includes an early version of TCP (encompassing the modern TCP and IP protocols)
- On **January 1, 1983** all hosts connected to ARPAnet were required to transition from NCP to TCP/IP



First Internet demonstration
(ARPAnet, PRNET, SATNET) in 1977

Vint Cerf

Robert Kahn

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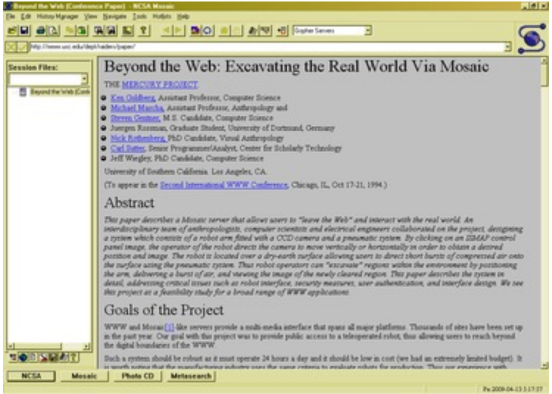
1991 – Creation of the World Wide Web (WWW) at CERN

- Between 1989 and 1991 Tim Berners Lee developed HTML, HTTP and an early browser and web server
- Belgian computer scientist Robert Cailliau contributed in the early stages

Robert Cailliau, Jean-François Abramatic and Tim Berners-Lee



Mosaic web browser

1990-2000 – An explosion of “killer” applications on the Internet

The Web

Email

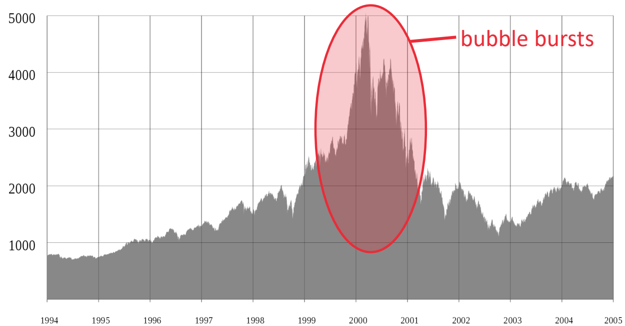
P2P filesharing

Instant messaging

2001 – Dotcom bubble bursts

- 1000s of Internet start-ups made public offerings
- Valued in the billions without revenue streams
- Stocks collapsed in 2000-2001
- A few "big winners" survived the crash
    - Microsoft
    - Cisco
    - Yahoo
    - eBay
    - Google
    - Amazon
    - ...

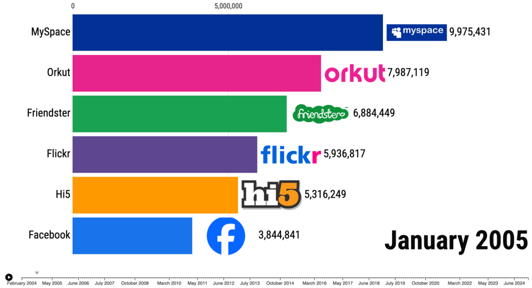
NASDAQ index between 1994 and 2005

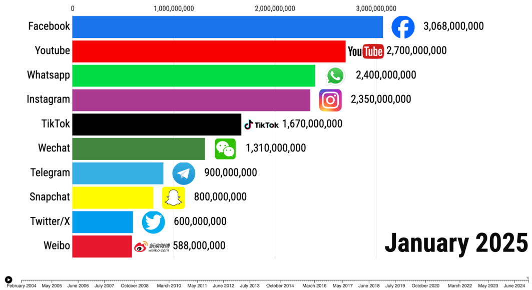


2000s – Birth of the interactive and social web

- **2003**: Launch of MySpace
  - Becomes most visited website on the web by 2005
  - Reaches peak of 115 million unique users in 2008
- **2006**: Facebook becomes accessible to the general public
  - Surpasses MySpace as most visited website by 2008
  - 430 million monthly active users in 2010
- Explosion of social networking websites and apps

…



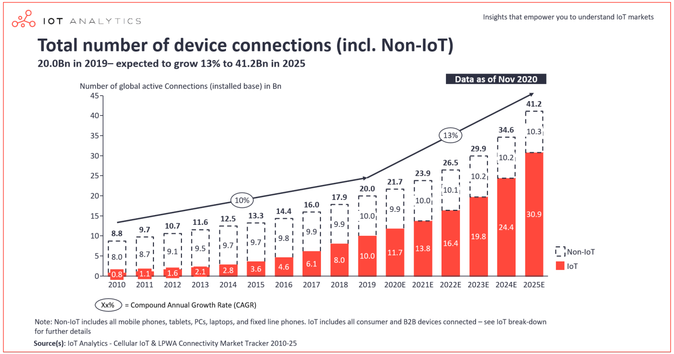


Source: *https://statisticsanddata.org/data/most-popular-social-media-2004-2025/*

2010s – The Internet of Things (IoT)

- Connecting not only people, but also “things” to the Internet
- Early experiments with Internet-connected devices in the 1990s
  - 1990: “Internet Toaster”
  - 1991: “Trojan Room Coffee Pot”

- In the **2010s** everything became connected to the Internet



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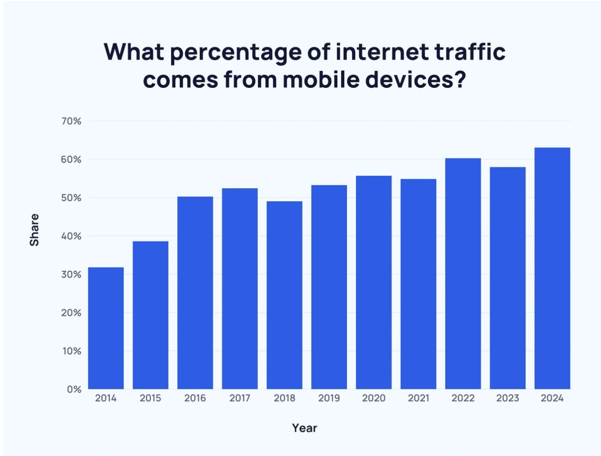
Source: *https://iot-analytics.com/state-of-the-iot-2020-12-billion-iot-connections-surpassing-non-iot-for-the-first-time/*

2010s – The mobile Internet era

- **2007**: Launch of the first iPhone
- Focus on mobile devices and experiences

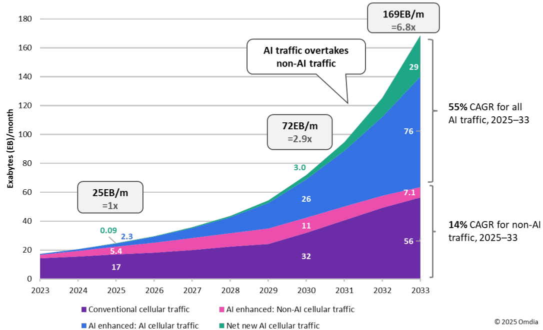
- Users can access the Internet anywhere at any time
- Widespread use of Internet streaming applications



Source: *https://explodingtopics.com/blog/mobile-internet-traffic*

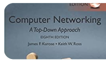
2020s – AI-native web

- Shift in how the Internet is being used
  - Before: Search → Click → Read → Decide
  - Now: Ask AI → Synthesized answer → Take action
- **2022**: Launch of ChatGPT to the public
- **2025**: Wide availability of Cloud-based AI services
- Possible **risks** for the future of the web / Internet
  - Flooding of the web with AI-generated content (e.g., "Dead Internet" theory)
  - Unauthorized use of (personal) data for AI model training
  - Need for massive compute and network resources (and energy), raising questions about sustainability



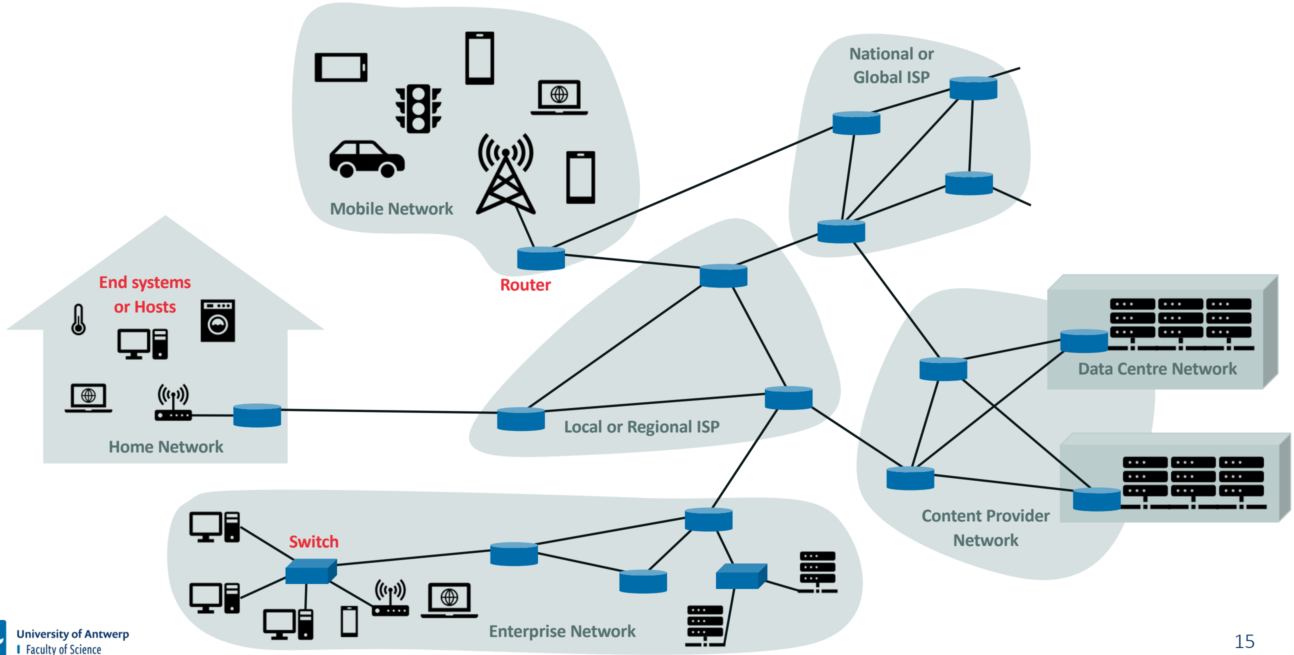
Source: *https://omdia.tech.informa.com/om135791/ai-network-traffic-report-update-for-europe-projected-cellular-growth-will-stress-access-infrastructure*

Internet Architecture

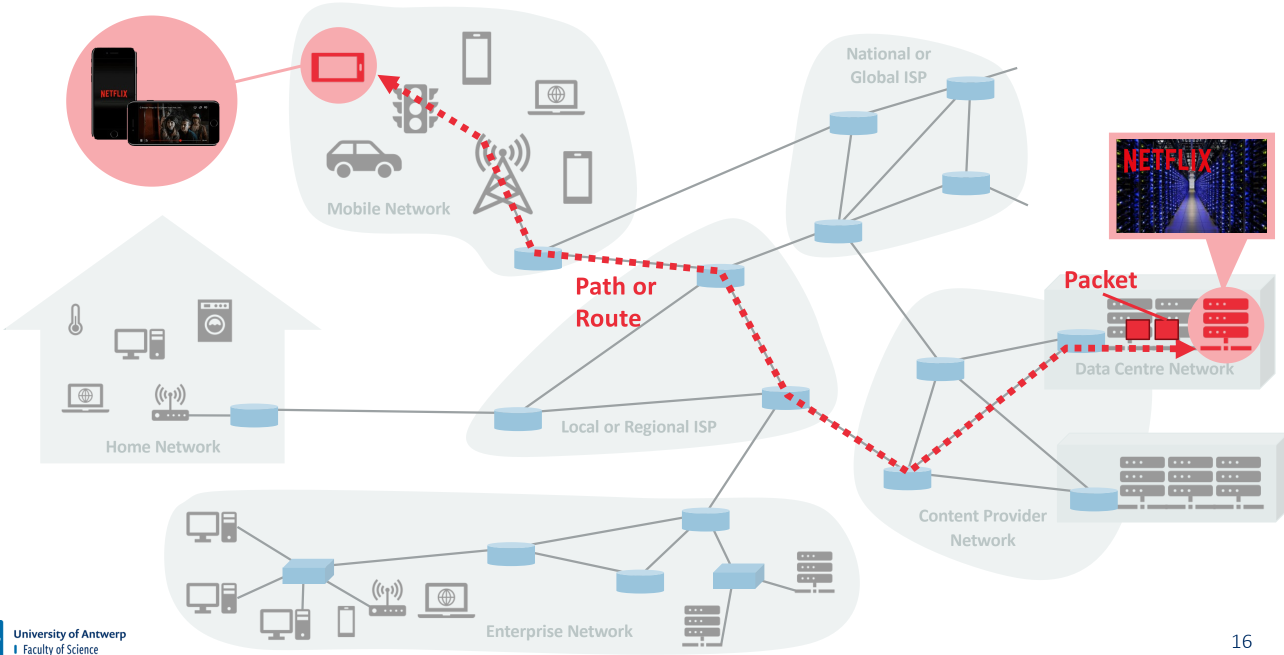


Sections 1.1 – 1.3

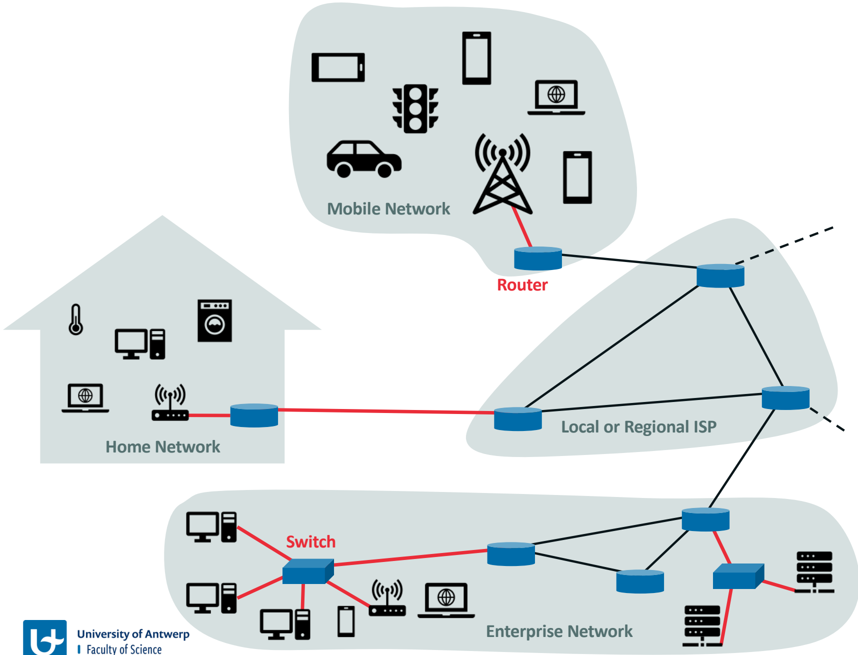
A high-level view of the Internet's architecture



Example: Watching movies online

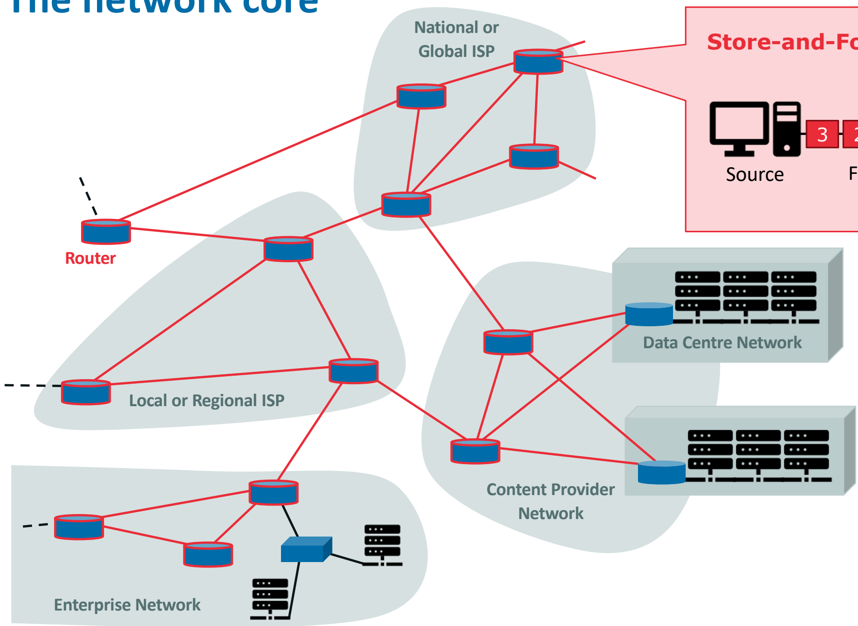


Access Networks

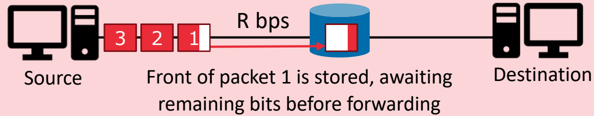


- **Home/enterprise access**
  - Digital Subscriber Line (DSL)
  - Cable
  - Fibre-to-the-Home (FTTH)
  - 5G Fixed Wireless
- **Access inside the home/enterprise**
  - Ethernet
  - Wireless LAN (e.g., Wi-Fi)
- **Wide-area wireless access**
  - 3G
  - 4G (LTE)
  - 5G (New Radio – NR)

The network core



Store-and-Forward Transmission



Packet transmission time

- Packet size: *L* bits
- Link transmission rate: *R* bps
- Transmission time from source to router: L / R seconds
- Transmission time from router to destination: L / R seconds
- Total transmission time:
  2 L / R seconds

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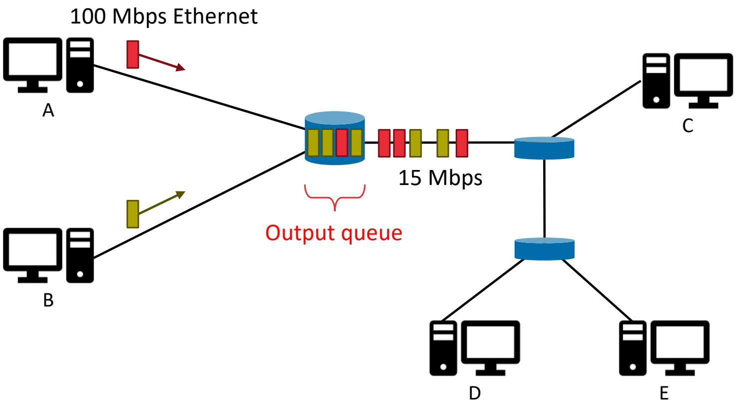
Packet switching



Output buffer (or queue)

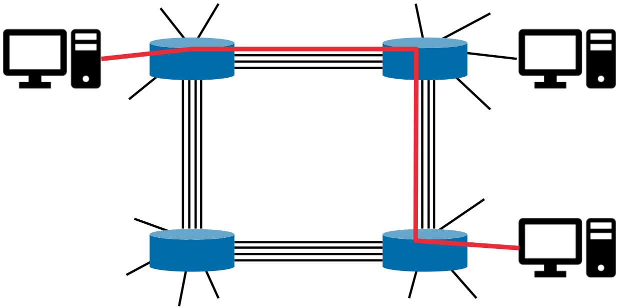
- Temporarily stores packets while associated link is busy
- Causes **queuing delays**
- If a packet arrives to a full buffer, **packet loss** occurs

Forwarding tables & routing protocols

- Determine how packets are forwarded by routers
- Each host is identified by an IP address
- **Forwarding tables** map destination IP to output links
- **Routing protocols** configure the forwarding tables

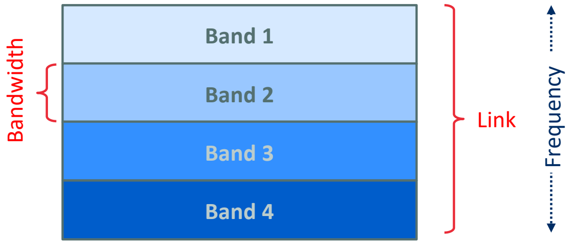
Circuit switching

- Alternative to packet switching
- Resources needed along the path (buffers, link transmission rate) to provide communication between the end systems are **reserved** for the duration of the communication session
- Example: Traditional telephone networks



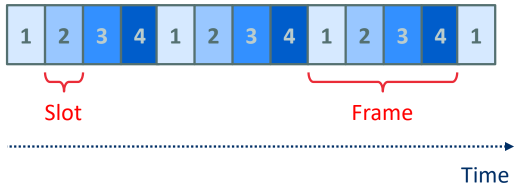
Multiplexing multiple circuits over one link

Frequency Division Multiplexing (FDM)



Each frequency band is dedicated to a specific sender-receiver pair

Time Division Multiplexing (TDM)



All slots with the same label are dedicated to a specific sender-receiver pair

Is packet switching or circuit switching better?

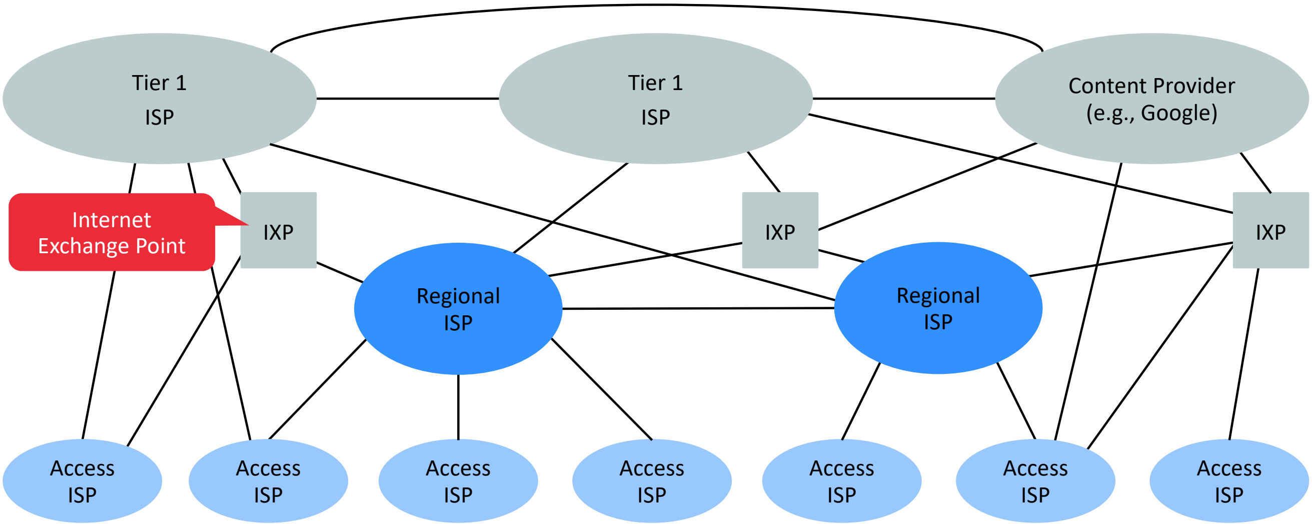
Packet switching is better, as it offers better capacity sharing

Packet switching is better, as it is simpler and cheaper to implement

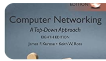
Circuit switching is better, as it provides more predictable end-to-end delays for real-time services

They both have their advantages and disadvantages

The Internet: A network of networks



Network Performance



Given a packet size of L bits, and transmission rate of R bps, how long would it take to transmit 3 packets from source to destination with 1 intermediary router?

2 L / R seconds

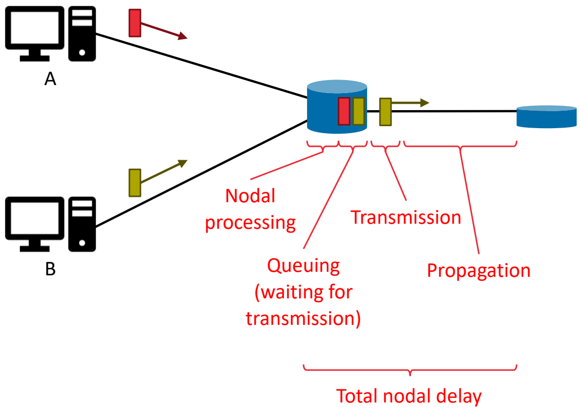
3 L / R seconds

4 L / R seconds

5 L / R seconds

6 L / R seconds

Types of nodal delay



- **Processing delay**
  - The time required to examine the packet's header and determine where to direct the packet
- **Queuing delay**
  - The time the packet spends waiting in the buffer to be transmitted
  - Depends on the number of earlier-arriving packets waiting
- **Transmission delay**
  - The time to transmit the packet on the link
  - Depends on the packet size L and transmission rate R
- **Propagation delay**
  - The time required for bits to propagate from the source to destination router of the link (limited by speed of light)
  - Propagation speed *s* is usually between 2 and 3 x 10^8 m/s
  - Propagation delay equals d/s, with *d* the link distance

End-to-end delay

- Packets accumulate nodal delay from all routers they pass on the path from source to destination
- Assuming a path with N – 1 routers between source and destination

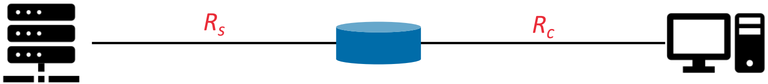
$$d_{e2e} = \sum_{n=1}^{N} d_{n,proc} + d_{n,queue} + d_{n,trans} + d_{n,prop}$$

- End-to-end delay can be analysed using the Traceroute programme
  - Sends 3 packets to each intermediary router, and determines delay to that router based on the response
  - Standardized as IETF RFC 1393

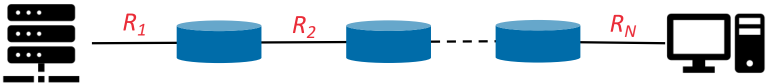
```
> tracert www.google.be

Tracing route to www.google.be [2a00:1450:400e:801::2003]
over a maximum of 30 hops:

  1     2 ms     1 ms     1 ms  ptr-g10bpkxncsi2klp2fmu.18120a2.ip6.access.telenet.be [2a02:1812:162d:6a00:5667:51ff:fedd:a8a6]
  2    27 ms    15 ms    13 ms  ptr-377whsjxcbe4fbcig8ap.18120a2.ip6.access.telenet.be [2a02:181f:0:80a3::1]
  3     *        *        *     Request timed out.
...
 13    37 ms     *        *     2001:4860:0:1::4fbd
 14    30 ms    40 ms    32 ms  2001:4860:0:1::4fc7
 15    60 ms    35 ms    34 ms  ams17s10-in-x03.1e100.net [2a00:1450:400e:801::2003]

Trace complete.
```

Throughput

- **Instantaneous throughput**: The rate (in bits/sec) at which a host is receiving data at a specific time
- **Average throughput**: The average rate (in bits/sec) at which an entire file is downloaded
  - If a file consists of *F* bits and the transfer takes *T* seconds, the average throughput is *F/T* bits/sec
- **Bottleneck link**: The link on the path from source to destination with the lowest transmission rate

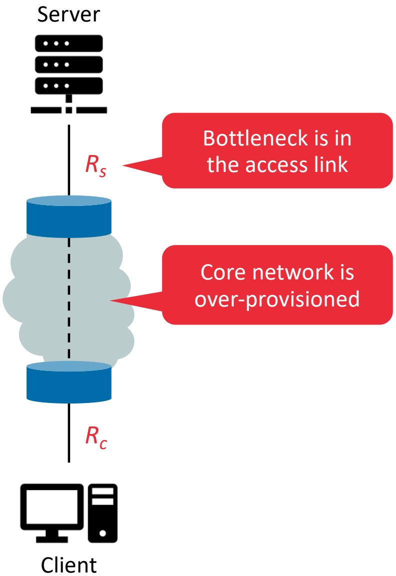


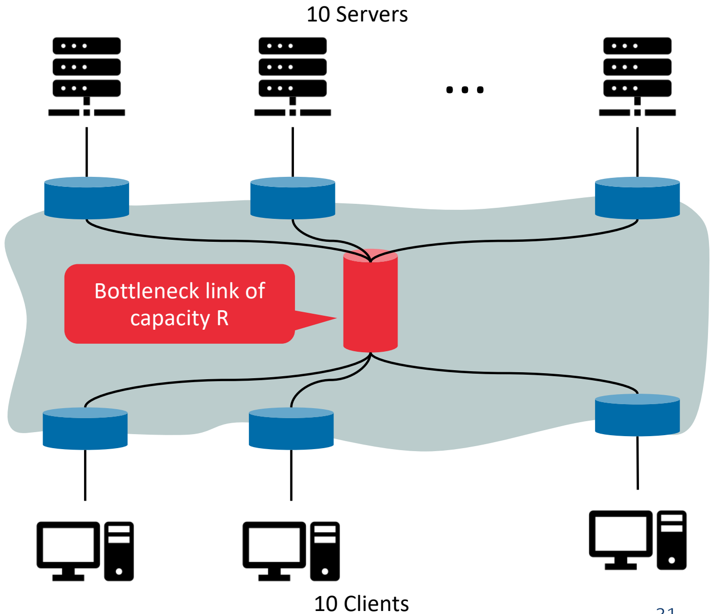
Throughput = $\min(R_s, R_c)$



Throughput = $\min(R_1, R_2, \ldots, R_N)$

Throughput depends on transmission rate and co-existing traffic

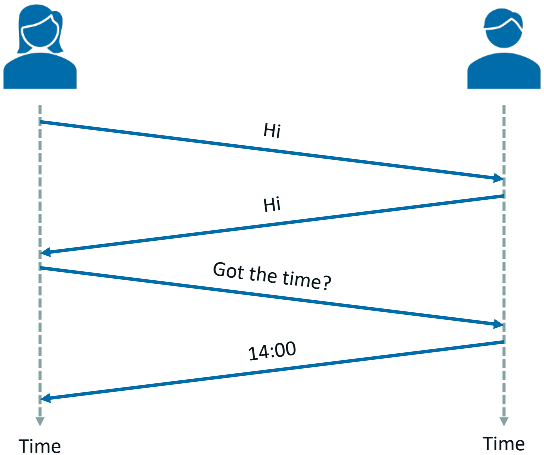




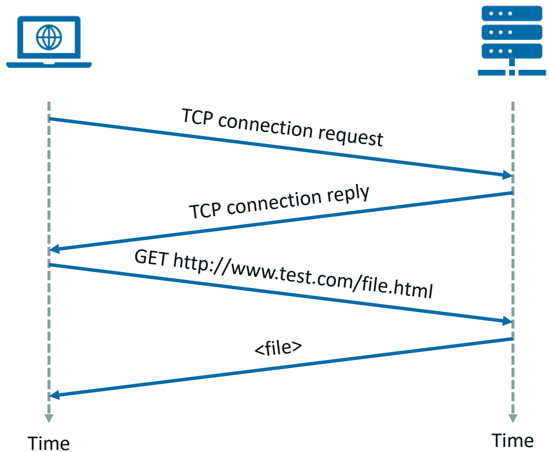
Protocols

Protocols define the conventions of communication between entities

Human protocol



Computer network protocol

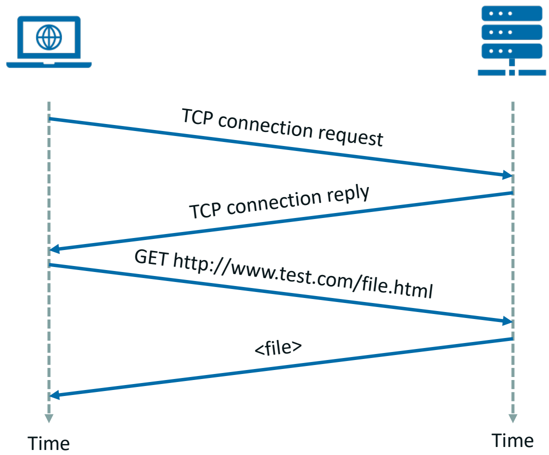


Definition of a computer network protocol

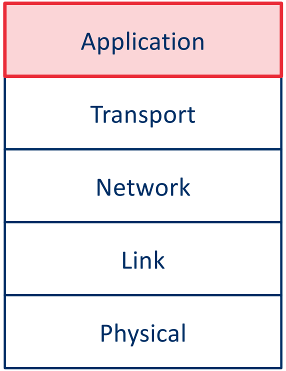
- "A protocol defines the **format and the order of messages exchanged** between two or more communicating entities, as well as the **actions taken** on the transmission and/or receipt of a message or other event."

- Standardization of network protocols
  - Internet Engineering Task Force (IETF)
    - Standards documents called Requests for Comments (RFCs)
    - TCP, IP, HTTP, TLS, IPSec, UDP, BGP, OSPF, …
  - Institute of Electrical and Electronics Engineers (IEEE)
    - IEEE 802.11 (Wi-Fi), IEEE 802.3 (Ethernet), …
  - 3rd Generation Partnership Project (3GPP)
    - 1G, 2G, 3G, 4G (LTE), 5G, …
  - ETSI
    - Network management, network virtualization, …

Computer network protocol



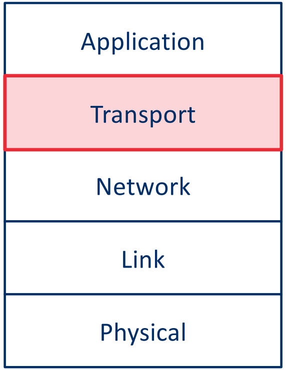
The Internet protocol stack consists of 5 layers



Application Layer

- Distributed over multiple end-systems
- Consists of the applications and their associated application-layer protocols
- Examples
  - HyperText Transfer Protocol (HTTP)
  - Simple Mail Transfer Protocol (SMTP)
  - File Transfer Protocol (FTP)
  - Domain Name System (DNS)
  - ...
- Packet of information is referred to as a **message** at this layer

The Internet protocol stack consists of 5 layers



Transport Layer

- Transports application-layer message between end-systems
- Transport Control Protocol (TCP)
  - Connection-oriented
  - Reliable (retransmits lost packets)
  - Flow control (sender/receiver speed matching)
  - Congestion control (throttles transmission rate to avoid network congestion)
- User Datagram Protocol (UDP)
  - Connection-less
  - Unreliable, no flow control, no congestion control
- Transport-layer packets are referred to as **segments**

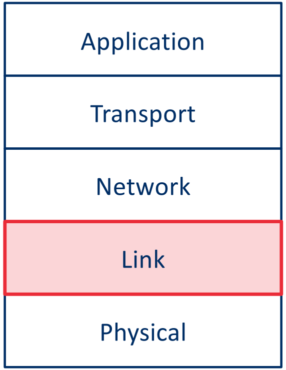
The Internet protocol stack consists of 5 layers

Application
Transport
Network
Link
Physical

Network Layer

- Responsible for moving network-layer packet from one host to another
- Functionality
  - Addressing (Internet Protocol (IP) versions 4 and 6)
  - Routing (many intra- and inter-domain routing protocols)
- Network-layer packets are called **datagrams**

The Internet protocol stack consists of 5 layers



Link Layer

- Responsible for transferring data across a single link (i.e., between 2 nodes)
- Functionality depends on the specific link-layer protocol
  - Reliability
  - Error detection
  - Addressing
  - Media access
  - …
- Examples
  - Ethernet
  - Wi-Fi
  - DOCSIS (Data Over Cable Service Interface Specification)
  - PPP (Point-to-Point Protocol)
- Link-layer packets are called **frames**

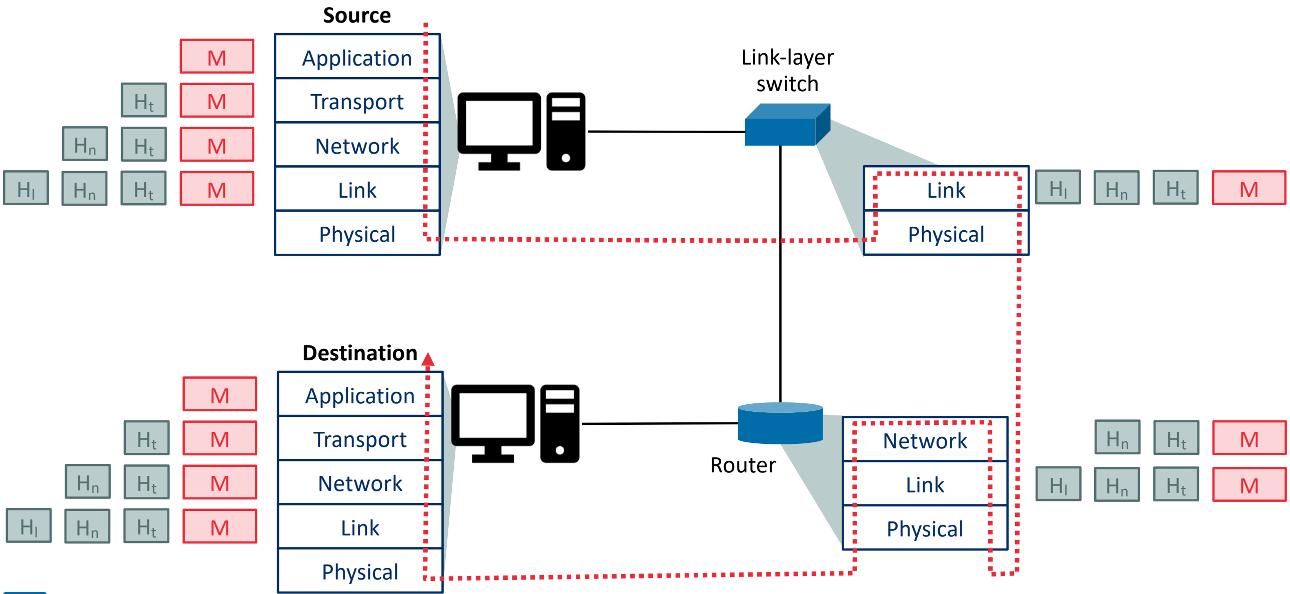
The Internet protocol stack consists of 5 layers

Application
Transport
Network
Link
Physical

Physical Layer

- Responsible for transferring individual symbols (one or more bits) over the physical medium
- Functionality
  - Modulation & coding
  - Bit synchronization
  - …
- Depends on the type of medium
  - Twisted-pair copper wire
  - Single-mode fiber optics
  - Free-space optical waves
  - Radio frequency waves
  - …

Encapsulation



Summary

Summary

- The Internet evolved from a network of 4 computers in 1967 to a **network of networks** containing billions of devices
- It consists of a layered hierarchy of **Internet Service Providers** (ISPs), together forming a global network connecting all hosts
- It mostly makes use of **packet switching**, which is more efficient and simple, but less predictable than **circuit switching**
- Network **performance** is characterised in terms of delay, packet loss and throughput
- A **protocol** defines the format and order of messages exchanged between hosts, and actions taken upon receiving/transmitting these messages
- The Internet **protocol stack** consists of 5 layers (application, transport, network, link, and physical), each encapsulating messages with their own headers

- The contents of this part is covered in the **book** in Sections 1.1, 1.2, 1.3, 1.4, 1.5 and 1.7

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End of Part 1